

Koushik Roy

Graduate Research Assistant and Graduate Teaching Assistant at Oklahoma State University

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Work Experience

Jan 2024 - Present

Graduate Research Assistant

The School of Electrical and Computer Engineering, Oklahoma State University

- Conducting research on improving EEG-based emotion recognition through multi-source domain adaptation under the guidance of Dr. Qi Cheng.
- Exploring methods to address inter-subject variability in EEG signals using multiple statistical alignment criteria.
- Investigating adaptive strategies for selecting relevant source domains to mitigate negative transfer effects.
- Evaluating proposed approaches on diverse EEG datasets, including the large-scale LEMON dataset, for cross-subject generalization.
- Presented a poster on Comparative Study of EEG Data Processing in Various Domains and Models for Eye State Classification.

Fall 2024, Spring
2025

Graduate Teaching Assistant

Division of Engineering Technology, Oklahoma State University

- Served as a Graduate Teaching Assistant for ENSC 2613: Introduction to Electrical Science in Fall 2024, Spring 2025, and Fall 2025.
- Conducted lab sessions focused on basic electrical circuits, including theoretical concepts and practical applications.
- Assisted with grading assignments and exams, and proctored tests throughout the semester.
- Supported students in understanding foundational electrical engineering principles during lab and office hours.

Aug 2022 - Dec 2023

Data Analyst

FundedNext, NEXT Ventures

- Processing, analyzing, visualizing data, and creating dashboards.
- Modularizing and automating data processing and reporting process.
- Applying machine learning for classification, regression, and forecasting problems.
- Tech Stack: R, Python, R Shiny Dashboard, AWS EC2 etc.

Apr 2022 - Jul 2022

Machine Learning Engineer

NybSys Ltd.

- Driver's attention monitoring and alert system.
- Masked face detection and recognition for facial authentication based security system.
- Hand written digit sequence recognition for optical mark recognition (OMR).
- Multi camera surveillance and monitoring solution for industrial usage.
- Tech Stack: Python, PyTorch, TensorFlow, OpenCV, Pandas, NumPy, MediaPipe etc.

Sep 2021 - Mar 2022

Data Scientist

NybSys Ltd.

- Location-based efficient navigation-equipped web app for emergency responders.
- Mixed fidelity synthetic data generation by combining statistical parameters of both low-fi and hi-fi data.
- Interactive web app for stock price prediction.
- Tech Stack: Python, R, Shiny, Leaflet, Dplyr, AWS, revgeo, ggmap, osrm, geosphere etc.

Education

2024 - Present	Oklahoma State University (Stillwater, Oklahoma) <i>PhD in Electrical and Computer Engineering</i> (Major: Electrical Engineering , GPA: 4.00/4.00 till Fall 25)
2022 - 2023	United International University (Dhaka, Bangladesh) <i>M.Sc. in Computer Science and Engineering</i> (Major: Data Science , CGPA: 3.78/4.00)
2017 - 2021	Military Institute of Science and Technology, Bangladesh University of Professionals (Dhaka, Bangladesh) <i>B.Sc. in Electrical, Electronic and Communication Engineering</i> (Major: Communication , CGPA: 3.64/4.00)

Research Interests

- Signal Processing • Deep Learning • Machine Learning • Domain Adaptation • Transfer Learning

Relevant Courses

- Computer Vision • Time Series • Stochastic Systems • Deep Learning • Robotics • Neural Network • Digital Signal Processing

Technical Skills & Certifications

- **Coding Skills:** *Python, R, SQL, Matlab, C*
- **Deep Learning & Computer-vision Packages:** *TensorFlow, PyTorch, Scikit-Learn, Open-CV*
- **Data Processing, Visualization & Deployment:** *R: tidyverse, dplyr, Plotly, Shiny Python: pandas, numpy, gradio*
- **Robotics:** *Arduino, Raspberry Pi, PCB Designing, Sensors & Micro-controllers*
- **Selected Certifications:** *TensorFlow Developer Certificate, Convolutional Neural Networks in TensorFlow, Crash Course on Python, Artificial Intelligence Foundations: Machine Learning, Intro to Deep Learning*

Academic Research Experience

Jan 2023 – September 2023	A master’s thesis on Adaptive dimensionality reduction technique for big data analytics <i>Supervisor: Dr. Dewan Md. Farid, Professor, Department of Computer Science and Engineering, United International University</i>
Feb 2020 – Feb 2021	An undergraduate thesis on Demand Forecasting in Smart Grid Using Long Short-term Memory (LSTM) <i>Supervisor: Dr. Kazi Abu Taher, Professor, Department of ICT, Bangladesh University of Professionals</i>

Publications

- 2025 | **Koushik Roy**, Qi Cheng, and Md. Mahmudul Hasan. Enhanced Multi-Source Domain Adaptation for EEG Pattern Recognition via Source Selection and Distance Function Optimization. *Asilomar Conference on Signals, Systems, and Computers (Asilomar)*, Pacific Grove, CA, USA, (publication processing).
- 2024 | **Koushik Roy** and Dewan Md Farid. An adaptive feature selection algorithm for student performance prediction. *IEEE Access*, 12:75577 – 75598, May 2024.
- 2023 | **Koushik Roy**, Huu-Hoa Nguyen, and Dewan Md Farid. Impact of dimensionality reduction techniques on student performance prediction using machine learning. *CTU Journal of Innovation and Sustainable Development*, 15:93–101, October 2023.
- 2022 | M. A. Hoque Akif, **K. Roy**, N. Abdullah, N. Z. Priota and M. Saif Hassan Onim, "Performance Analysis of Machine Learning Models for Cheating Detection in Online Examinations," 2022 25th International Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh, 2022, pp. 342-347, doi: 10.1109/ICCIT57492.2022.10055801.
- A. Ishmam, M. Hasan, M. Hassan Onim, **K. Roy**, M. Hoque Akif and H. Nyeem, "Modelling Lips State Detection Using CNN for Non-verbal Communications", *Lecture Notes in Networks and Systems*, pp. 59-70, 2022. doi: 10.1007/978-981-19-2445-3_5
- M. S. Onim, H. Nyeem, **K. Roy**, M. Hasan, A. Ishmam, M. A. Akif, and T. B. Ovi, "BLPnet: A new DNN model and Bengali OCR engine for automatic licence plate recognition," *Array*, vol. 15, p. 100244, 2022, doi: 10.1016/j.array.2022.100244
- K. Roy** and M. A. H. Akif, "Real Time Hand Gesture Based User Friendly Human Computer Interaction System," 2022 International Conference on Innovations in Science, Engineering and Technology (ICISSET), 2022, pp. 260-265, doi: 10.1109/ICISSET54810.2022.9775918
- K. Roy** and D. Chanda, "A Robust Webcam-based Eye Gaze Estimation System for Human-Computer Interaction," 2022 International Conference on Innovations in Science, Engineering and Technology (ICISSET), 2022, pp. 146-151, doi: 10.1109/ICISSET54810.2022.9775896
- 2021 | **K. Roy**, A. Ishmam and K. A. Taher, "Demand Forecasting in Smart Grid Using Long Short-Term Memory," 2021 International Conference on Automation, Control and Mechatronics for Industry 4.0 (ACMI), 2021, pp. 1-5, doi: 10.1109/ACMI53878.2021.9528277

Awards

08/19/2024 - | **Leo J. and Josie Moseley Peters Scholarship**
12/13/2024 | *Oklahoma State University*